

LISHI, China

WMO: 537640

Lat: 37.5067N

Lon: 111.1132E

Elev: 3120

StdP: 13.11

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.1	6.7	-14.5	2.8	13.7	-10.6	3.5	15.9	15.1	26.3	13.4	25.5	4.7	20	0.438

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	92.1	66.8	89.6	65.7	87.2	65.4	73.8	83.5	72.1	81.7	70.5	79.9	6.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.1	129.0	79.1	69.3	121.0	77.4	67.5	113.7	76.0	39.9	83.5	38.2	81.8	36.6	79.7	85.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.5	13.1	11.4	DB	-2.3	96.5	4.9	2.9	-5.8	98.6	-8.7	100.2	-11.4	101.8	-14.9	103.9	(4)
			WB	-3.6	76.5	4.6	3.1	-6.9	78.8	-9.6	80.6	-12.1	82.3	-15.4	84.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	21.9	30.0	42.6	55.2	65.1	72.9	76.3	73.0	63.8	51.9	38.1	25.1
	DBStd	19.78	7.07	7.99	7.90	7.99	5.99	4.71	4.29	4.75	5.96	6.35	7.13	6.86
	HDD50	2908	871	559	251	39	1	0	0	0	1	55	359	772
	HDD65	5940	1336	979	694	307	74	4	0	4	90	407	808	1237
	CDD50	3427	0	0	22	194	467	687	815	712	415	113	2	0
	CDD65	985	0	0	0	12	76	241	350	251	54	1	0	0
	CDH74	9264	0	0	8	221	980	2488	3185	1925	440	17	0	0
	CDH80	3368	0	0	1	53	270	996	1318	622	107	1	0	0
Wind	WSAvg	4.8	4.7	4.9	5.4	5.7	5.6	5.1	4.5	4.3	4.2	4.5	4.5	4.5
Precipitation	PrecAvg	19.7	0.1	0.2	0.4	0.9	1.5	2.3	4.6	4.5	3.1	1.3	0.6	0.1
	PrecMax	29.4	0.4	0.9	1.3	2.3	6.3	5.4	10.9	10.5	8.1	4.8	2.0	0.6
	PrecMin	9.8	0.0	0.0	0.0	0.1	0.1	0.3	1.1	1.1	0.6	0.2	0.0	0.0
	PrecStd	5.1	0.1	0.2	0.3	0.6	1.3	1.4	2.6	2.4	2.0	1.1	0.6	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.1	58.2	73.7	86.4	90.1	95.3	96.2	91.7	89.1	76.3	63.1	47.7
		MCWB	35.0	41.2	49.7	56.3	60.7	64.0	69.3	70.0	65.0	57.7	47.4	35.5
	2%	DB	41.8	51.3	66.8	79.9	86.1	91.2	92.8	88.8	82.7	71.3	58.3	42.9
		MCWB	31.6	37.9	46.8	54.3	58.9	62.9	68.5	68.7	63.1	54.4	44.9	33.4
	5%	DB	37.7	47.3	61.9	75.4	83.1	88.5	90.2	86.3	79.0	67.5	54.4	39.5
		MCWB	29.0	35.7	44.3	51.9	57.4	62.5	68.3	68.2	62.3	53.2	43.0	30.9
	10%	DB	34.0	43.8	57.3	71.0	79.9	85.9	87.4	83.7	75.4	63.8	50.5	36.1
		MCWB	26.8	33.7	41.9	50.1	56.2	62.0	68.2	67.6	60.8	51.4	41.5	29.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.0	42.6	52.0	61.1	64.7	70.9	76.5	75.7	70.5	59.9	49.9	37.5
		MCDB	45.1	53.7	69.8	75.9	82.1	84.9	87.7	84.3	79.8	69.3	57.6	44.8
	2%	WB	32.0	39.4	48.2	57.2	62.0	68.4	74.6	73.4	67.2	57.5	47.2	34.0
		MCDB	40.5	48.9	63.2	73.7	77.4	81.8	85.3	81.9	76.8	66.3	54.4	40.7
	5%	WB	29.6	36.7	45.5	54.4	60.4	66.7	72.9	71.8	64.7	55.5	44.9	32.0
		MCDB	36.8	45.6	59.6	70.0	76.6	79.5	82.9	80.8	74.0	63.6	52.5	37.8
	10%	WB	27.2	34.4	42.9	52.1	58.6	65.2	71.3	70.3	62.7	53.6	42.1	30.0
		MCDB	33.4	42.6	55.6	66.5	74.1	78.4	81.1	79.0	71.9	61.4	49.8	35.9
Mean Daily Temperature Range	5% DB	MDBR	19.9	20.6	21.8	23.0	23.3	22.0	18.5	17.7	18.4	19.6	19.1	18.1
		MCDBR	25.9	26.2	29.9	30.2	30.1	27.9	24.4	22.3	24.8	26.1	25.9	22.9
		MCWBR	17.8	15.8	15.4	12.9	11.9	9.4	7.7	7.6	9.4	12.1	14.7	15.1
	5% WB	MCDBR	24.1	23.3	27.5	24.6	23.3	20.7	18.2	17.0	19.4	20.4	22.1	20.2
		MCWBR	16.9	14.4	15.1	11.7	10.2	8.5	7.6	7.3	8.9	10.5	13.5	13.8
Clear-Sky Solar Irradiance	taub	0.395	0.486	0.510	0.540	0.567	0.643	0.629	0.636	0.573	0.530	0.421	0.378	
	taud	1.985	1.795	1.784	1.758	1.765	1.659	1.743	1.709	1.790	1.816	1.977	2.048	
	Ebn at Noon	235	226	236	239	235	217	219	213	217	210	224	232	
	Edh at Noon	42	58	64	70	71	79	72	73	63	56	42	37	
All-Sky Solar Radiation	RadAvg	882	1093	1502	1792	1964	1873	1712	1554	1298	1127	924	801	
	RadStd	87	118	127	146	141	143	141	110	118	126	125	81	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.65	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page